Natural Language Processing

Autumn 2026

Week 1: Subject Introduction and Preparation

Weekly lectures and workshops

- This subject consists of weekly online lectures and in-person workshops.
- You need to select a workshop time on your timetable to be able to do the Quiz (assignment 2)
- It is important to come to the workshops. It gives you a better learning experience and helps you to form groups for Assignment 3.

Contents

- **Introduction**
  - NLP
  - Testing current NLP products
  - Why NLP is hard
  - First step of NLP: Tokenization

- **Overview of weekly topics**
- **Assessments**
- **Google Colab Preparation**

What is NLP?

Natural Language Processing (NLP) is a sub-field of artificial intelligence that helps computers understand, interpret and manipulate human language.

“Natural language processing (NLP) is a subfield of **computer science** and especially **artificial intelligence**. It is primarily concerned with providing computers with the ability to process data **encoded** in **natural language** and is thus closely related to information retrieval, knowledge representation and computational linguistics” (Wikipedia)

Then vs. Now:

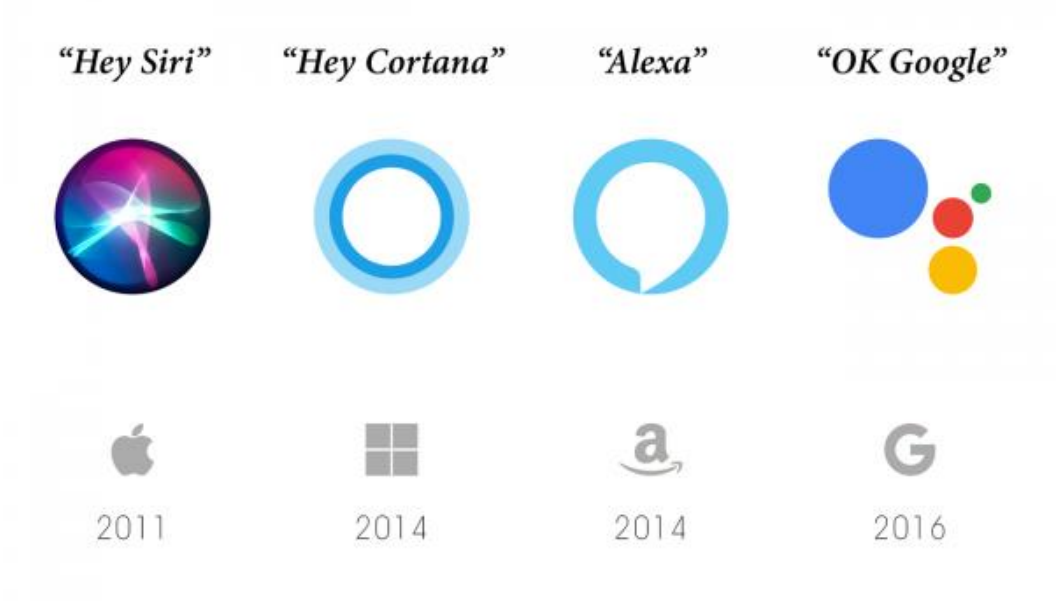
•*Then:* Rule-based systems, specific tasks.

•*Now:* Large Language Models (LLMs), reasoning, and generation.

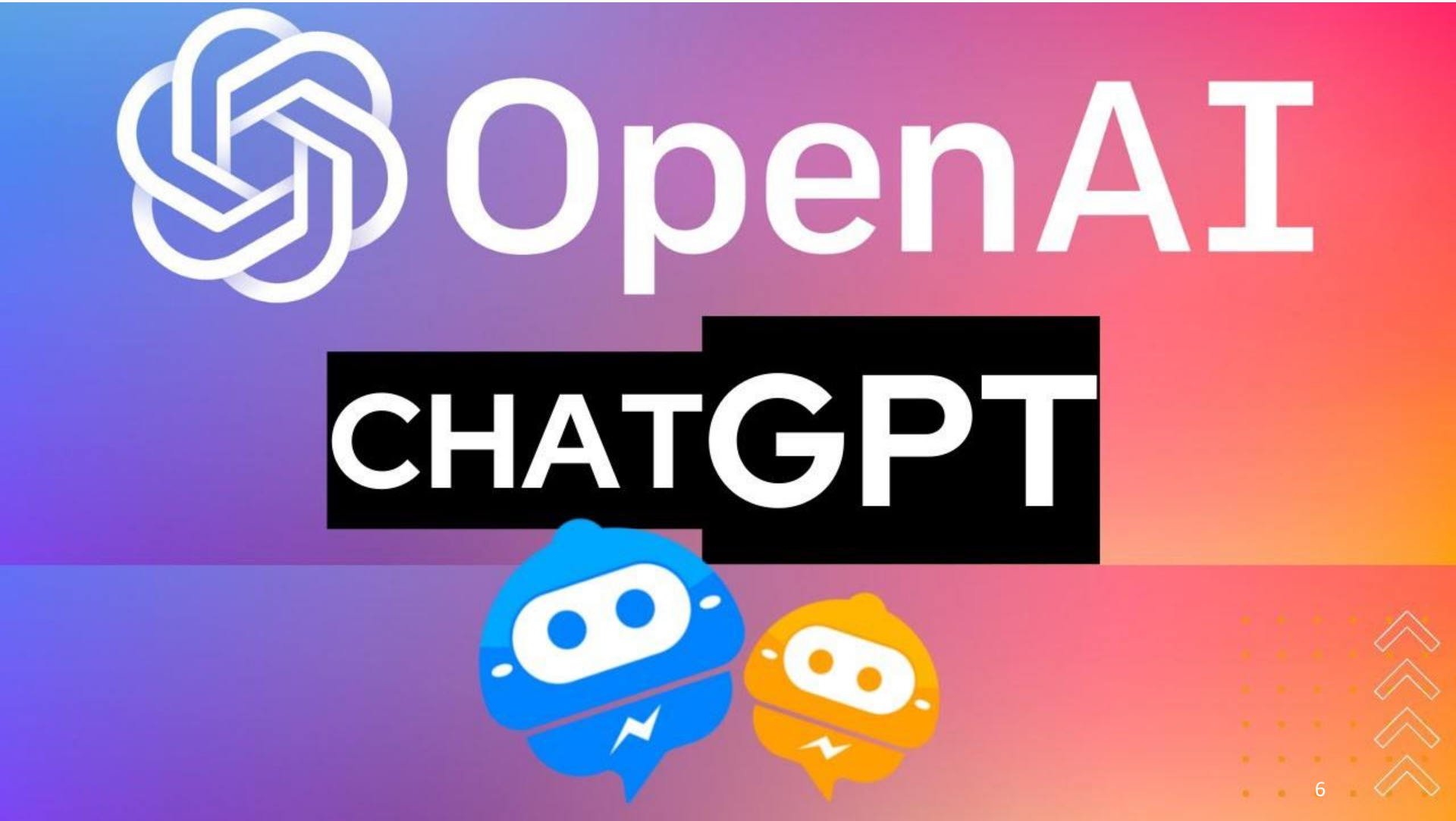
NLP is everywhere – though probably unnoticed

Goal: for computers to process or "understand" natural language in order to perform tasks that are useful, e.g.,

- Performing Tasks, like making appointments, buying things
- Question Answering - Siri, Google Assistant, Facebook Messenger, Cortana …



That’s all until - ChatGPT



Is it Human-like intelligence?

- Given the amazing performance of AI, do you think LLMs underlying ChatGPT understand human language like humans do? Why?

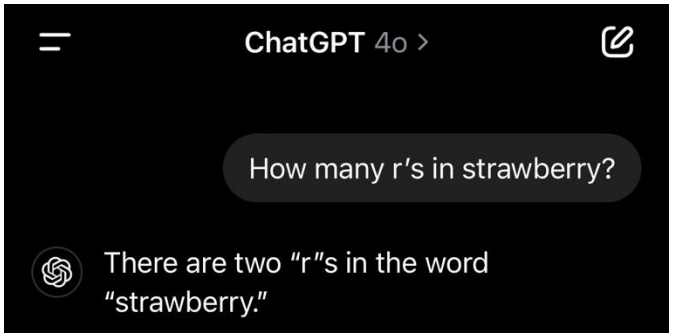
- Let's watch this video ("Reading Pigeon") from the 1950s:

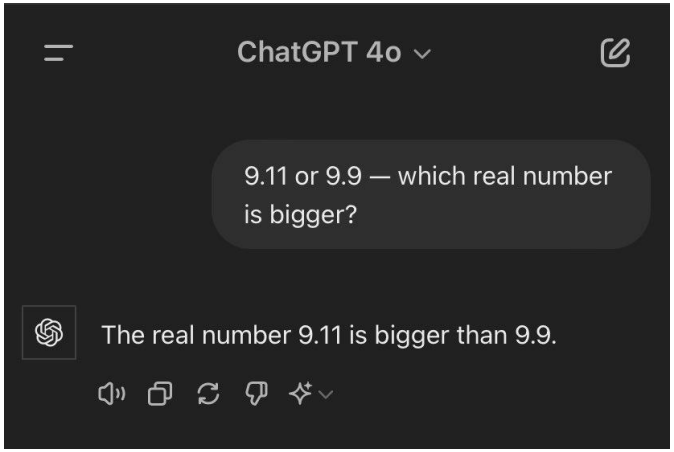
https://youtu.be/qTXGAd1kpXY?si=I42rK5Nqj6OXbxnl

- Do you think LLMs may be behaving like the pigeon?

How reliable are LLMs? (1/3)

- In 2024:





How reliable are LLMs? (2/3)

- In 2025:

ChatGPT 4o

How many r there are in the word strawberry

There are **three** "r" letters in the word **strawberry**.

How many letters of symmetry there are in this word?

In the word **strawberry**, there are **no letters of symmetry** if we consider vertical or horizontal symmetry.

A **letter of symmetry** is a letter that looks the same when reflected along a vertical or horizontal axis.

- **Vertical symmetry**: Letters like A, H, I, M, O, T, U, V, W, X, and Y have vertical symmetry.
- **Horizontal symmetry**: Letters like B, C, D, E, H, I, K, O, and X have horizontal symmetry.

Checking the letters in **strawberry**:

- **S, T, R, A, W, B, E, R, R, Y**
- **Only "A" has vertical symmetry**, but no letter in the word has perfect horizontal symmetry.

So, the answer is **one letter of symmetry: "A" (vertical symmetry).**

How reliable are LLMs? (3/3)

- In 2026:

I'm going to wash my car. The car wash shop is 20 meters away. Should I walk there or drive there?

20 meters? 😊
That's about **20–25 seconds of walking**… basically your driveway with extra steps.

Walk 👍

Pros

- Faster overall (no starting engine, no reversing, no parking)
- No fuel burned (even a cold start uses noticeable fuel)
- Better for the engine (short drives are actually *bad* — engine never warms up → condensation → wear)
- Free mini-exercise

Drive 👎

Cons

- Takes longer than walking
- Worst possible trip length for a car mechanically
- Looks a bit… dramatic

Verdict:
Walk.
Driving 20 m is mechanically worse for the car than leaving it parked — you'd literally be harming the engine slightly to save ~15 seconds.

You'll arrive before the car's infotainment system even boots. 🚶

Why is NLP hard?

- Complexity in representing, learning and using linguistic/situational/world knowledge
- Human languages are ambiguous (unlike programming and other formal languages), and human language interpretation depends on real world, common sense, and contextual knowledge

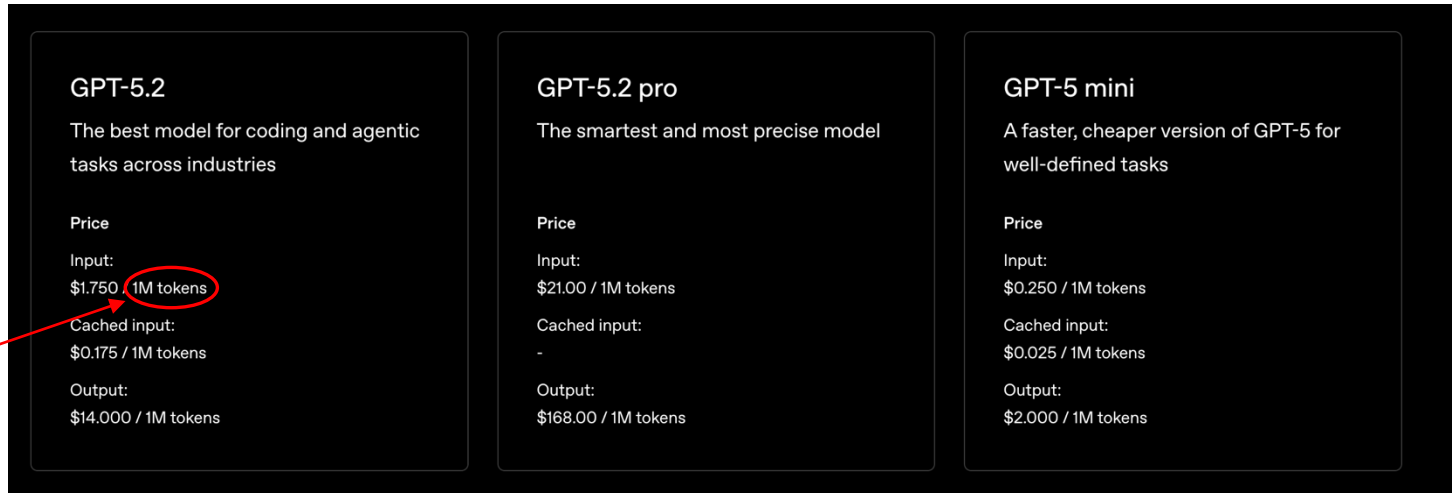
- Example: "I saw the dog with the telescope."
- Question: Who has the telescope?

- Another example: “The trophy doesn’t fit into the suitcase because it is too big.”
- Question: What is too big?

Tokenization

- Tokenization is a crucial first step in most NLP tasks.
- Tokenization is the process of breaking text into smaller units, called tokens. These tokens could be words, subwords, or characters.
- Example:
  - Input: "I love NLP!"
  - Output: ["I", "love", "NLP", "!"]

API Cost by Tokens



Types of Tokenization

- Word Tokenization: Splits text into individual words.
  - Example: "I love NLP!" → ["I", "love", "NLP", "!"]
- Sentence Tokenization: Splits text into sentences.
  - Example: "I love NLP. It's fun!" → ["I love NLP.", "It's fun!"]
- Subword Tokenization: Splits words into smaller units (e.g., prefixes, suffixes).
  - Example: "unhappiness" → ["un", "happiness"]
  - Helpful for out-of-vocabulary (OOV) words

Impacts of Tokenization

- **Text Classification** (Assigning predefined categories to a document).
  - Example: Classifying movie reviews as "positive" or "negative" based on the tokens (words) found in the review.

- **Machine Translation** (Translating text from one language to another).
  - Example: Translating the English sentence "How are you?" into French. Tokenization helps map each word to its equivalent in French, like "Comment" (How), "êtes" (are), and "vous" (you).

- **Text Generation and Language Modeling** (Generating human-like text or predicting the next word in a sequence)
  - Example: Predicting the next token in a sentence, such as "The weather is [sunny]" after tokenizing the input text.

Tokenization in Non-English Languages

- Tokenization in non-English languages presents several challenges due to linguistic characteristics that differ significantly from English.
- One particular example of such a challenge is the lack of word delimiters (space).
  - Languages like Chinese/Japanese/Thai do not use spaces to separate words. The phrase "我喜欢自然语言处理" (“I like natural language processing”) is a continuous string of characters without spaces.
- Specialized tokenizers are needed to process those languages.

Overview of this Subject

Week 1: Subject Introduction and Preparation

Week 2: Word Embedding

Week 3: POS Tagging

Week 4: Named-Entity Recognition

Week 5: Text CNNs

Week 6: RNN and LSTM

Week 7: Transformer and BERT

Week 8: Large Language Models

Week 9: Model Distillation

Week 10: AI Agents

Week 11: Adversarial Robustness of AI Models

Week 12: Project Presentations

Why are fundamental stuff (still) important to know?

- Should we (AI experts) be “Drivers” or “Mechanics”?

A Car Driver (GenAI User)	**A Car Mechanic (AI Expert)**
Uses the car to get from A to B.	Understands the engine, torque, and fuel.
Experience: High-level, intuitive, "Black Box."	**Experience:** Low-level, structural, "White Box."
Limitation: If the car breaks, they are stranded.	**Capability:** Can tune, repair, and improve a car when needed.

We need to be “Mechanics”

- We (AI experts) should be the “Mechanics”.
- Everyone can talk to a machine. However, only an AI/NLP expert can explain why the machine talked back.
  - When an LLM hallucinations, you can't just "ask it why." Understanding Word Embeddings allows you to see the mathematical distance between concepts and fix biases at the source.
  - You don't use a canon to kill an ant. For tasks like PoS Tagging or NER, a specialized small model is 100x faster, cheaper, and more private than a massive LLM.
  - The "Attention" mechanism used by GPT was born from the evolution of embeddings and tagging. To build the next LLM, you must understand the DNA of the current ones.

Word Embedding (next week)

- How can we represent words as numbers so that computers can process them?

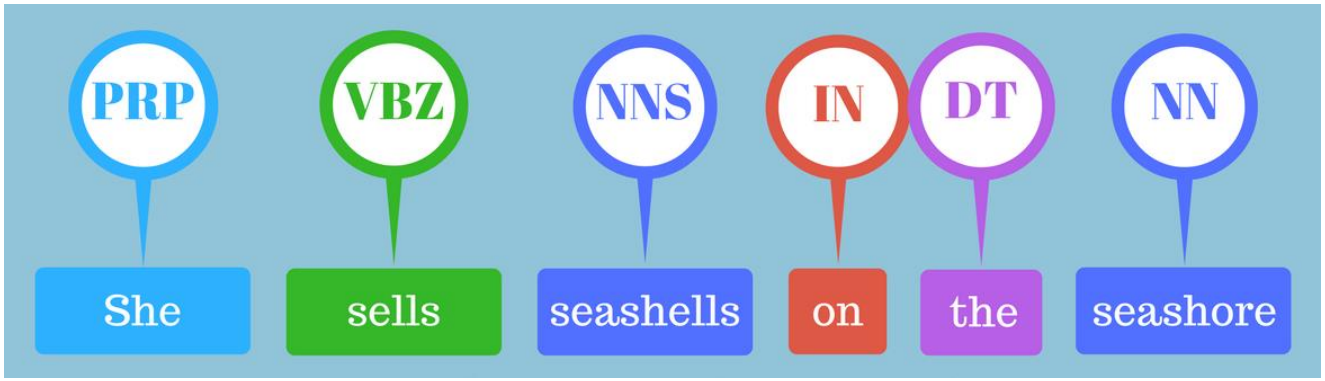
A 4-dimensional embedding

cat =>	1.2	-0.1	4.3	3.2
mat =>	0.4	2.5	-0.9	0.5
on =>	2.1	0.3	0.1	0.4
...	...			

Image Credit: TensorFlow

POS Tagging (Week 3)

- How can we assign words to their respective parts of speech to facilitate downstream applications, like grammar checking?



Named-Entity Recognition (Week 4)

- How can we identify predefined categories of objects in a body of text to facilitate application tasks, like financial analysis, where we want to find mentions on stocks?

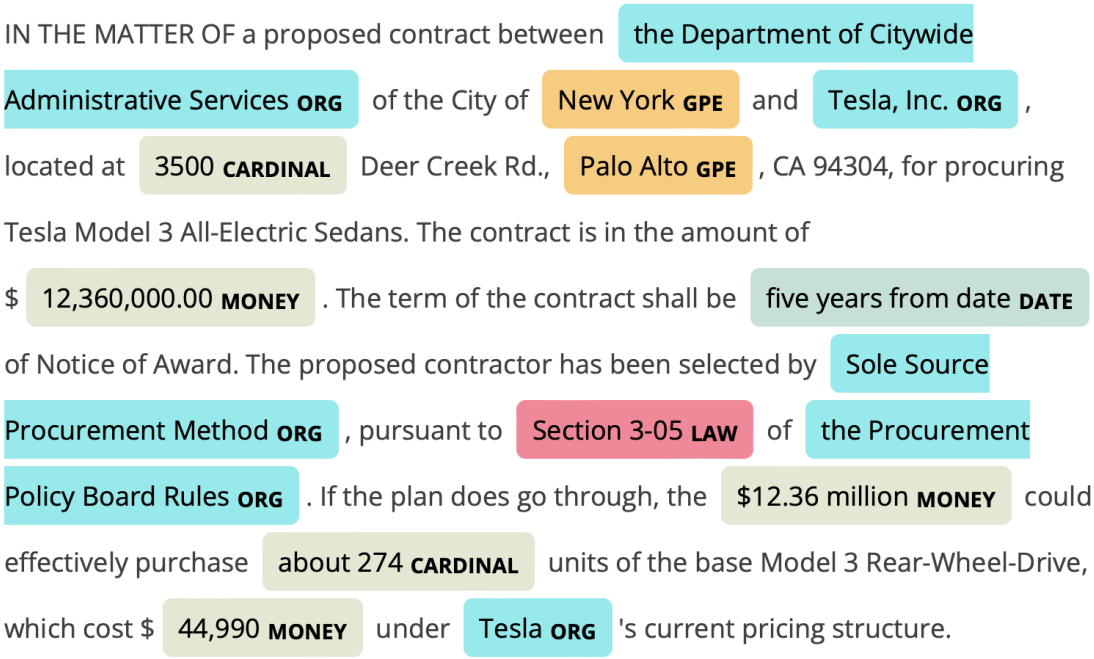


Image: https://dataknowsall.com/blog/ner.html

Text CNNs (Week 5)

- How can we use Convolutional Neural Networks (CNNs) to process text data?

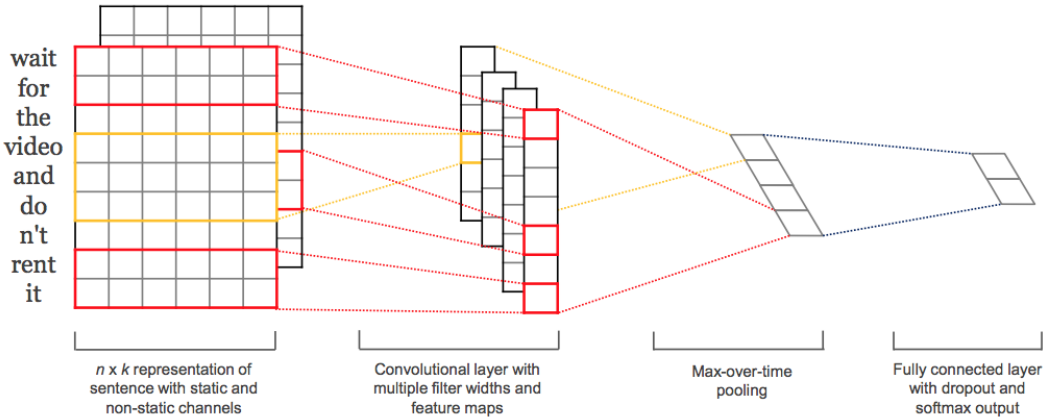


Figure 1: Model architecture with two channels for an example sentence.

Image: https://arxiv.org/pdf/1408.5882

Recurrent neural networks (RNNs) (Week 6)

- How can we use RNNs to process text data and convert it into another sequence, like translation?

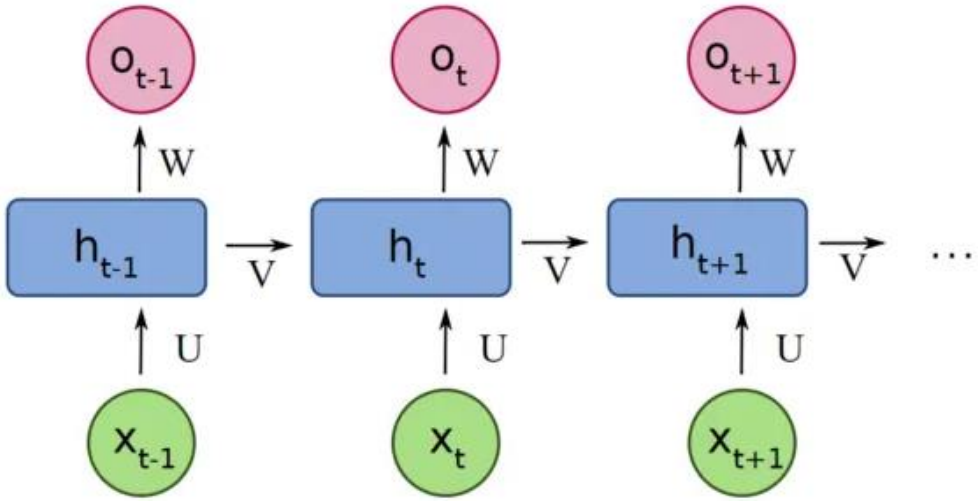


Image: https://tinyurl.com/ye93vh7u

Transformer (Week 7)

- Almost all currently state-of-the-art AI models use Transformer architectures.

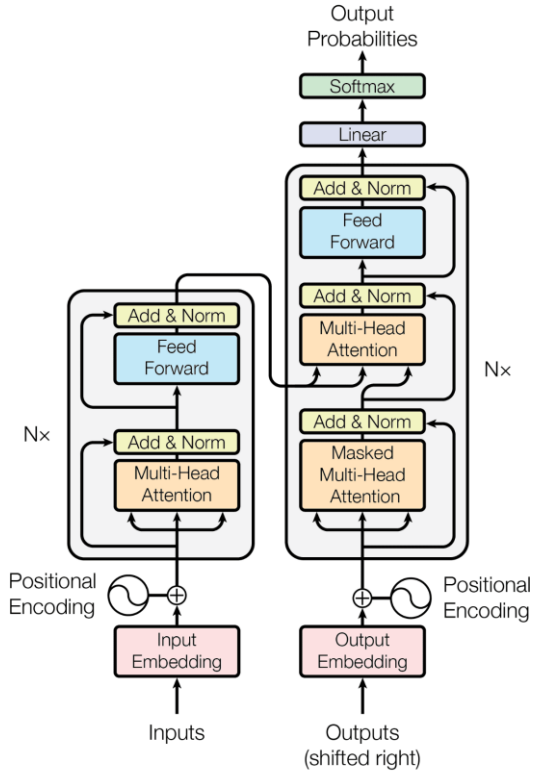


Image: https://arxiv.org/pdf/1706.03762

Large Language Models (LLMs) (Week 8)

- What are LLMs? What is prompt engineering, and Retrieval-Augmented Generation (RAG)?

Model Distillation (Week 9)

- How can we transfer knowledge from a large model to a smaller model?

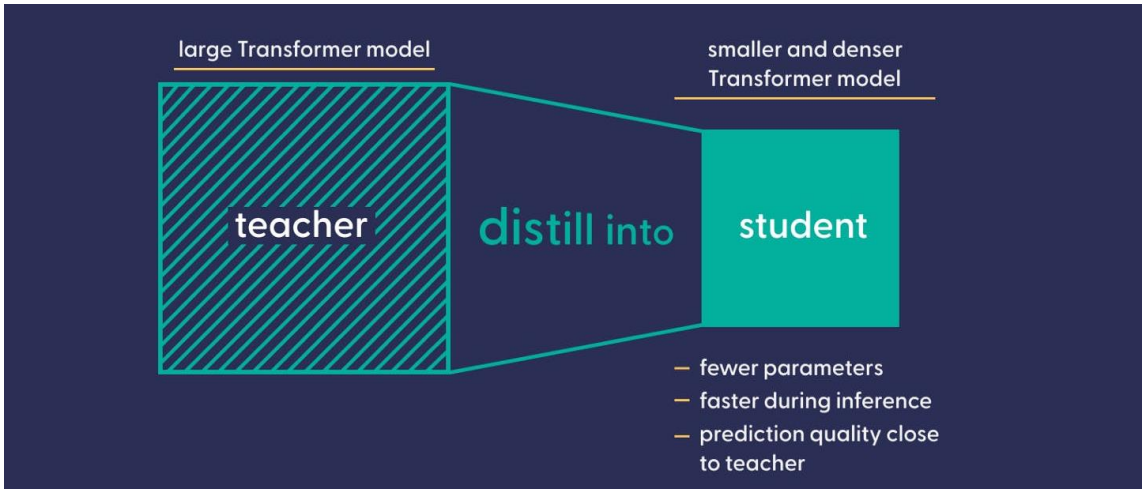


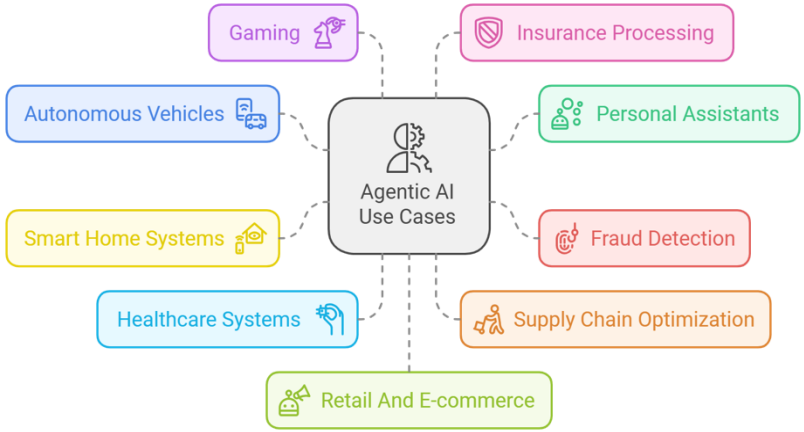
Image: https://tinyurl.com/37cz8cn6

AI Agents (Week 10)

AI agents transform how we interact with machines, such as enabling intelligent assistants, automated research tools, and more.

Key Features:

- Sets and manages its own goals
- Can plan multi-step actions
- Adapts to changes in the environment
- Can use external tools



Images: https://www.ampcome.com/post/top-20-agentic-ai-use-cases-with-real-life-examples

Adversarial Robustness of AI Models (Week 11)

- How do we assess the ability of an AI model to resist being tricked or exploited?

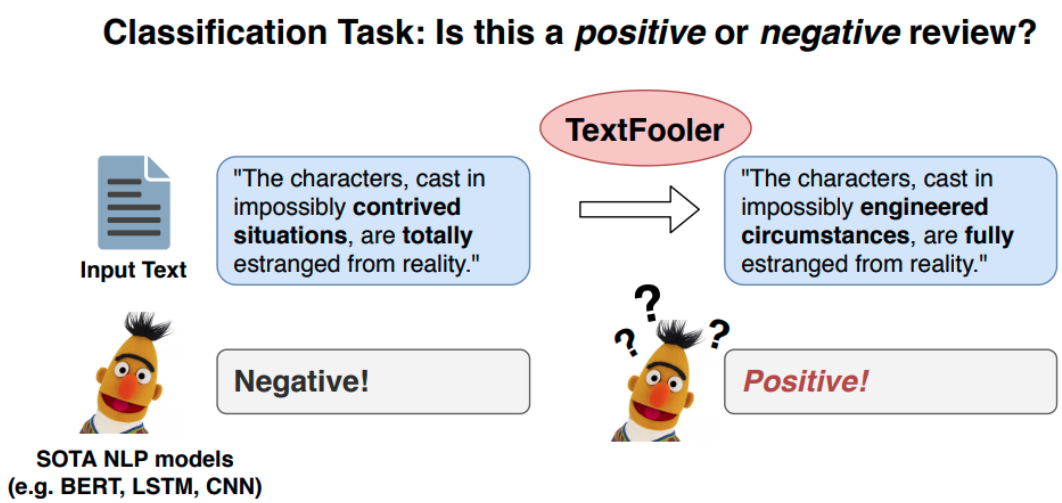


Image: https://arxiv.org/pdf/1907.11932

Assessment Overview

- Assessment 1 (Individual) [20 marks] Due 13 Mar at 11:59 pm
  - Search and Analyse NLP-related jobs and projects
  - Think about what skills and knowledge you need to develop
  - A stepping stone for your assessment 3

- Assessment 2 (individual) [30 marks] In Weeks 6 and 10 During the First 30 Min of Workshops
  - Open-book and Multiple-choice questions only
  - You can only attend the session shown on your Timetable to do the quiz

- Assessment 3 (Group with peer assessment) [50 marks] Presentation in Week 12's workshops, report due 15 May at 11:59 pm
  - Up to five people in a group
  - Propose and solve an application problem
  - Describe your solution and explain how it solves the problem
  - Demonstrate your project in a presentation and a report

Can you use GenAI in this subject's assessments?

- You are allowed to use AI tools to assist you in Assessment 1 and 3 submissions. However, you must acknowledge when and how you have used information or content generated from or revised by AI tools.
- You are not allowed to use AI tools for Assessment 2 (Quiz).

30

Python Basics

- This is not a Python class, but some basic Python understanding is needed to read the code.

- Self-paced Python learning materials are available freely online:
  - Google's Python Class.
  - W3Schools Python Tutorial.
  - Python for Beginners - Learn Coding with Python in 1 Hour.

Google Colab Preparation

- You will need a Google account
- Our weekly exercises are designed to run on the free version of Colab, utilising small models and small datasets (or a fraction of the original datasets). You can apply the same approach in your Assessment 3.
- "Hello, world" in Colab: https://colab.research.google.com/drive/1oU2VycL_FMRsvv1EBwFkzcThqVPGCKff?usp=sharing

Questions?